

**INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR**

- 1. Name of the Academic Unit: Metallurgical and Materials Engineering**
- 2. Subject Name: Extractive Metallurgy      L-T-P: 3-0-0      Credits: 3**
- 3. Pre-requisites: TRANSPORT PHENOMENA IN METALLURGICAL PROCESSES**
- 4. Syllabus and reference books:**

**Syllabus:**

**Introduction: Classification of Metals and Extraction Overview**

**Mineral dressing: comminution, sizing, classification, gravity separation, magnetic and electrostatic separation, flotation, ore sorting; solid-liquid separation.**

**Fuels for metallurgical processes; metallurgical furnaces; heat utilization in furnaces and heat exchangers; ore fines agglomeration; Furnace Refractories.**

**Unit Processes in pyrometallurgy: calcination, roasting, smelting, conversion; Thermodynamic principles of extraction.**

**Unit processes in hydrometallurgy: pretreatments, leaching, Purification of leach liquor, reduction and recovery of metals from aqueous solution.**

**Unit processes in electrometallurgy: Faradays laws of electrolysis, concept of overvoltage; electro-winning and electro-refining**

**Extraction and refining of metals: Cu, Ni, Pb, Zn etc. from sulfide ores; Fe, Al, Sn, Mg, ferroalloys etc. from oxide ores; Extraction of Ti and Zr through halide route.**

**Sustainable metal extraction: Recovery of by-product metals and treatment of metallurgical wastes; Metal recycling.**

**Reference Books:**

**1) *Physical chemistry of metallurgical processes*, M. Shamsuddin, 2016, The Minerals, Metals & Materials Society and John Wiley & Sons, Inc.**

**2) *Mineral Processing Technology*, 8th Ed., Barry A. Wills and James A. Finch, 2016, Elsevier Ltd.**

**3) *Extraction of Nonferrous Metals*, H. S. Ray, R. Sridhar, K. P. Abraham, Affiliated East-West Press Pvt. Ltd, Reprint 2018**

**4) *Treatise on Process Metallurgy*, Vol 3, 2014, Editor-in-Chief: SESHADRI SEETHARAMAN, Elsevier Ltd.**

**5) *Chemical Metallurgy: Principles and Practice*, Chiranjib Kumar Gupta, 2003, WILEY-VCH.**

### 5. Lecture-wise break-up:

| Sl. No.               | Topic                                                                                                                                                                           | No. of lectures |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.                    | Classification and uses of metals, Principles followed in metal extraction, Overview of the Routes of Metal extraction and Refining                                             | 2               |
| 2.                    | Types and Occurrence of Metal bearing Minerals, Mineral Beneficiation: Unit Steps, Evaluation of Mineral Beneficiation Operation                                                | 2               |
| 3.                    | Physical Separation of Minerals: Crushing and Grinding (Liberation), Screening and Classification, Solid-Liquid Separation                                                      | 2               |
| 4.                    | Physical Concentration of Minerals: Gravity Separation, Magnetic and Electrostatic Separation, Froth Flotation, Automated Sorting                                               | 3               |
| 5.                    | Classification and Use of Fuels in Metallurgical Processes, Characterization of Fuels, Coke-Making, Combustion and Flame Temperature                                            | 3               |
| 6.                    | Types of Metallurgical Furnaces, Fuel Utilization in Furnaces, Heat Exchanger, Furnace Refractories, Agglomeration of ores                                                      | 3               |
| 7.                    | Unit Processes for Chemical Treatment of Ores: Calcination, Roasting, Thermodynamics of Roasting: Predominant Area Diagram                                                      | 2               |
| 8.                    | Unit Process for Extraction from Oxides: Reduction Smelting, Thermodynamics Principles and Application in Iron, Zinc, Lead, Tin, Magnesium, Silicon etc. Extraction             | 4               |
| 9.                    | Unit Process for Extraction from Sulfides: Matte Smelting, Conversion, Application to Copper and Nickel Extraction, Flash Smelting, Continuous Conversion                       | 4               |
| 10.                   | Unit Process for Halide-metallurgy: Kroll Process; Refining of Titanium, Uranium, Silicon etc.                                                                                  | 2               |
| 11.                   | Refining of Metals: Physical and Chemical methods of Refining, Refining of Lead, Iron, Copper etc.                                                                              | 3               |
| 12.                   | Unit Process in Hydrometallurgy: Leaching, Solvent extraction, ion-exchange, adsorption on charcoal, reduction of ions in solutions, application in Gold, Nickel, Alumina       | 3               |
| 13.                   | Unit Process in Electrometallurgy: Laws of Electrolytic Cell Reduction and Refining, Current Efficiency and Power Requirements, Overvoltage, Application in Aluminum Extraction | 3               |
| 14.                   | Environmental Concern and Sustainable Metal Extraction, Modern Developments in Metal Extraction and Refining, Waste Recycling                                                   | 2               |
| Total number of hours |                                                                                                                                                                                 | 38              |